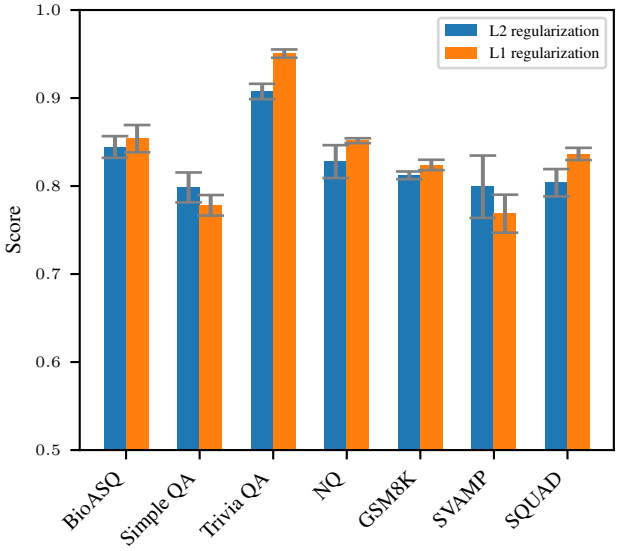

Figure 54. AUROC of linear probes at the stop token of the output, trained with either L1 or L2 regularization.

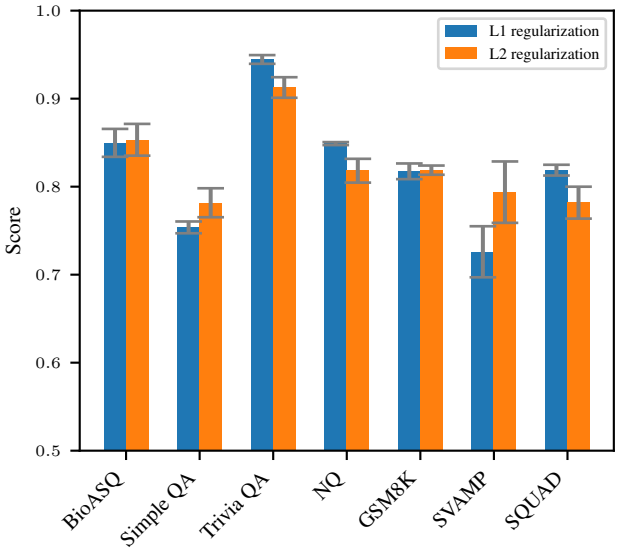

Figure 55. AUROC of linear probes at the token before the stop token of the output, trained with either L1 or L2 regularization.

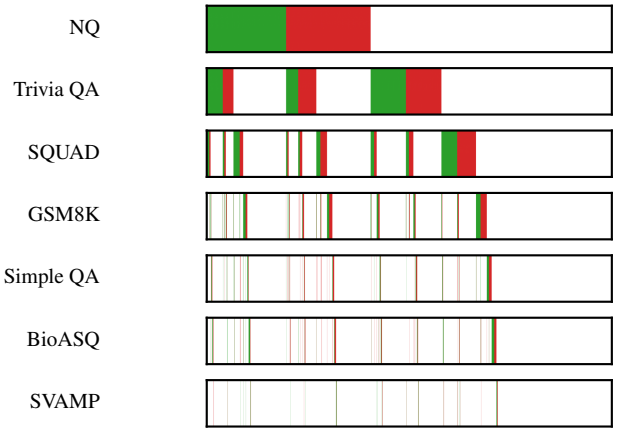

Figure 56. Signed support of sparse probes trained on different datasets at the stop token of the output, using L1 regularization at layer 32. Each row represents one probe trained on the corresponding dataset (y-axis labels). The x-axis shows the 3584 dimensions of the hidden state vector. Green indicates positive coefficients, red indicates negative coefficients, and white indicates zero coefficients (sparsity). Dimensions are sorted by sparsity level across all datasets, with the least sparse dimensions on the left and the most sparse on the right.

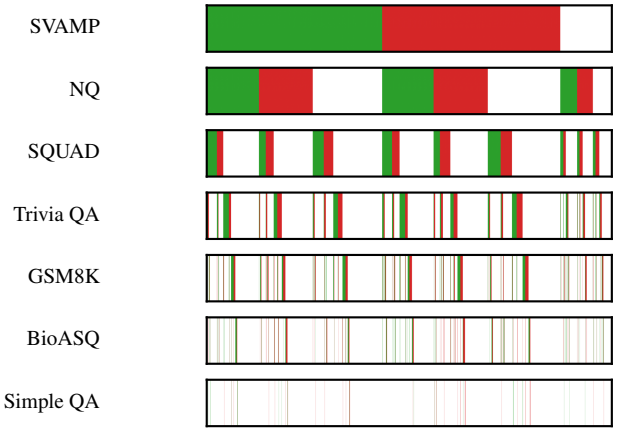

Figure 57. Signed support of sparse probes trained on different datasets at the token before the stop token of the output, using L1 regularization at layer 32. Each row represents one probe trained on the corresponding dataset (y-axis labels). The x-axis shows the 3584 dimensions of the hidden state vector. Green indicates positive coefficients, red indicates negative coefficients, and white indicates zero coefficients (sparsity). Dimensions are sorted by sparsity level across all datasets, with the least sparse dimensions on the left and the most sparse on the right.

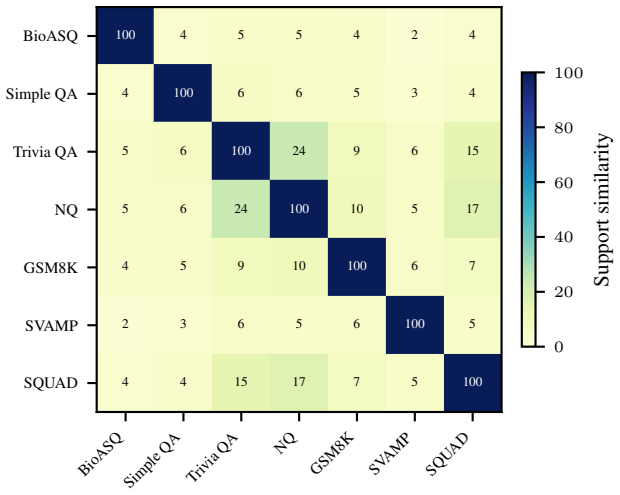

Figure 58. Support overlap between sparse probes trained on different datasets. Darker colors indicate higher overlap percentages. Task pairs with $> 30\%$ overlap (TriviaQA, NQ, SimpleQA) correspond to successful cross-task generalization, while most pairs show $< 15\%$ overlap, explaining generalization failure.

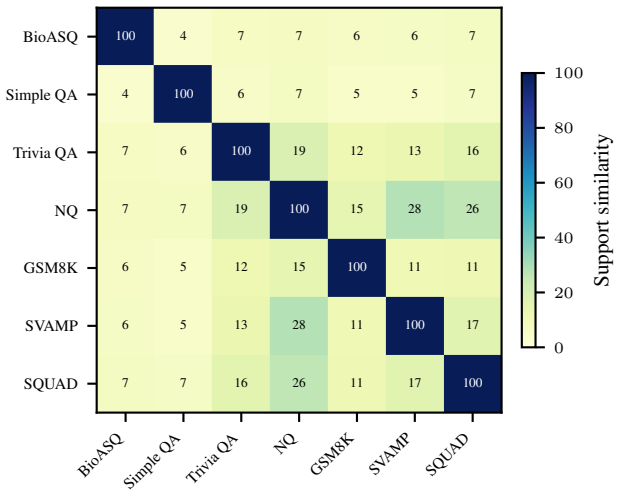

Figure 59. Support overlap between sparse probes trained on different datasets. Darker colors indicate higher overlap percentages. Task pairs with $> 30\%$ overlap (TriviaQA, NQ, SimpleQA) correspond to successful cross-task generalization, while most pairs show $< 15\%$ overlap, explaining generalization failure.

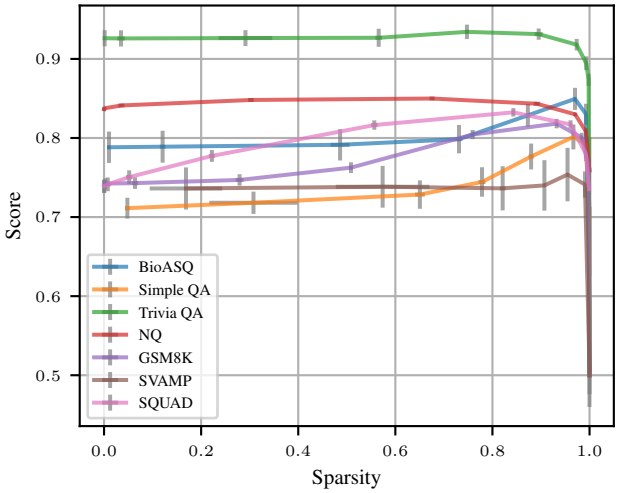

Figure 60. AUROC of probes trained using L1 regularisation as a function of the sparsity level on the stop token of the output.

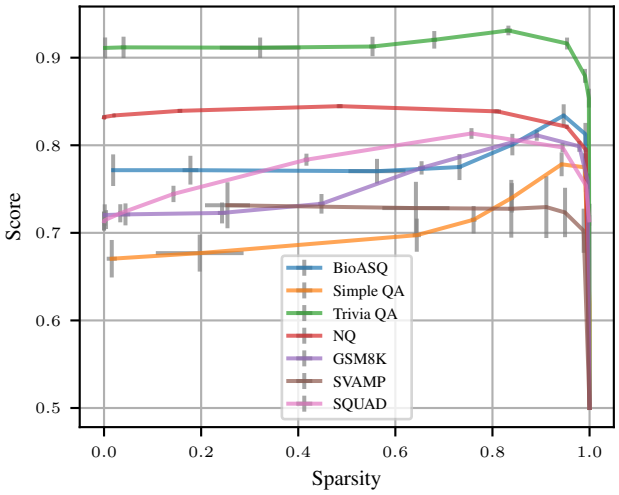

Figure 61. AUROC of probes trained using L1 regularisation as a function of the sparsity level on the token before the stop token of the output.

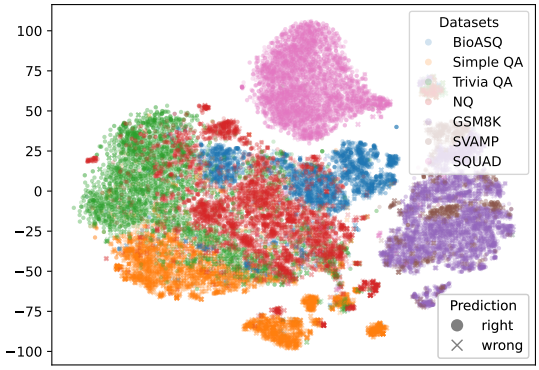

Figure 62. t-SNE plots of the hidden space at layer 32 at the stop token of the output.

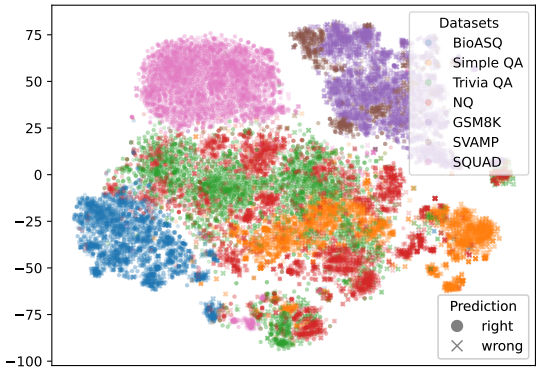

Figure 63. t-SNE plots of the hidden space at layer 32 at the token before stop token of the output.